NORTHEASTERN UNIVERSITY

COLLEGE OF ENGINEERING



Robotics Sensing & Navigation

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-Final Project Report-

Effects of Assisted GPS on Cold First Fix Localization Accuracy

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1. Background

When a GNSS receiver powers on without cached almanac or ephemeris data, it must perform a cold first fix, a full satellite search that can take 30 seconds to several minutes, with inaccurate position estimates throughout [1]. Assisted GPS (A-GPS) narrows this search by downloading satellite parameters over cellular or Wi-Fi, reducing time-to-first-fix (TTFF) from minutes to seconds, and may also improve early fix accuracy by enabling faster filter convergence [2]. However, the specific contribution of A-GPS to localization accuracy, as distinct from receiver hardware or proprietary fusion algorithms, remains poorly isolated in the literature [3]. To address this, we compared cold-start position error convergence across three systems under identical conditions: a smartphone with A-GPS enabled, the same smartphone in airplane mode (unassisted), and a standalone VectorNav VN-100 IMU + GlobalSat BU-353S4 GPS fused via a custom ROS2 EKF, to determine whether A-GPS improves accuracy, or only speed. We conduct this test in both stationary and driving conditions.

2. Objective and Methodology

The goal of this project is to quantify how Assisted GPS affects the accuracy and convergence rate of position estimates during and immediately after a cold first fix.

2.1 Experimental Systems

Four concurrent systems were operated during all trials:

- Phone (A-GPS ON): iPhone with cellular network connected, A-GPS enabled, and internal sensor fusion algorithms active.
- Phone (Unassisted): The same iPhone in airplane mode with GPS manually re-enabled. Identical hardware, no assistance data.
- Standalone Sensor Fusion (EKF): VectorNav VN-100 IMU at 40 Hz combined with a GlobalSat BU-353S4 GPS at 1 Hz. A custom Python EKF running on a laptop via ROS2 fused the two streams. No A-GPS ever.
- Ground Truth:
    - Driving: A second pre-converged iPhone (stable GPS lock for 10+ minutes before each trial) co-mounted on the vehicle, used as the position reference for all error calculations.
    - Stationary: the average location estimation over the full run.

2.2 Hardware and Data Collection

For stationary collection, all sensors were held within 5 meters of each other in an open, outdoor, unmoving venue (Centennial Common). For driving collection, all sensors were rigidly co-mounted on a personal vehicle to ensure identical motion across all systems. Cold starts were enforced by resetting location services on the experimental iPhone and power-cycling the standalone GPS at the start of each trial. The IMU was published on /imu at 40 Hz, and the GPS was published on /gps/fix at 1 Hz. Both iPhones logged position at 1 Hz using phyphox, with metadata time sheets providing Unix epoch timestamps for cross-device clock synchronization.

Two types of trials were conducted:

- Stationary: 10 trials (5 A-GPS, 5 unassisted), approximately 3 minutes each, between Centennial Common and Ryder Hall on the NEU Boston campus.
- Driving: 2 loop routes × 2 conditions (airplane mode on/off) = 4 trials, approximately 15 minutes each, near NEU campus around Ruggles station.

2.3 Custom ROS2 Package: agps_driver

We developed a custom ROS2 package, agps_driver, with three components:

- vn100_driver.py: Parses $VNYMR over serial at 115200 baud with XOR checksum validation. Euler→quaternion conversion. Publishes sensor_msgs/Imu on /imu at 40 Hz (yaw, pitch, roll, accel X/Y/Z, gyro X/Y/Z).
- gps_driver.py: Parses $GPGGA at 4800 baud, NMEA→decimal degrees. Publishes sensor_msgs/NavSatFix on /gps/fix at 1 Hz (lat, lon, alt, fix quality).
- agps_launch.py: Launches both nodes simultaneously with configurable port arguments.

2.4 Extended Kalman Filter

We implemented a five-state Extended Kalman Filter in Python to fuse the VN-100 IMU and BU-353S4 GPS into a single navigation solution. The state vector consists of east position, north position, east velocity, north velocity, and heading, all in a local East-North frame relative to the first valid GPS fix.

At each IMU reading (40 Hz), the filter predicts the new state by integrating the accelerometer (double integration for position, single for velocity) and gyroscope (single integration for heading). Before integration, the raw accelerometer values are corrected for gravity bias by subtracting the mean of the first 100 samples, collected while the sensor was stationary. The corrected body-frame accelerations are rotated into the East-North frame using the current heading estimate. The covariance is propagated as $P = FPF^T + Q$ with diagonal process noise values of 0.5 m^2/s (position), 5.0 m^2/s^3 (velocity), and 0.05 rad^2/s (heading).

At each GPS fix (1 Hz), the filter computes the difference between the GPS-reported position and the predicted position, then corrects the state using the Kalman gain. The GPS measurement noise is set to 9.0 m^2 (3 m standard deviation), reflecting open-sky BU-353S4 accuracy. Only position is observed, the BU-353S4 provides no velocity or heading.

The gravity calibration is critical: without it, the gravitational component in the accelerometer output causes hundreds of meters of drift when double-integrated over a five-minute trial.

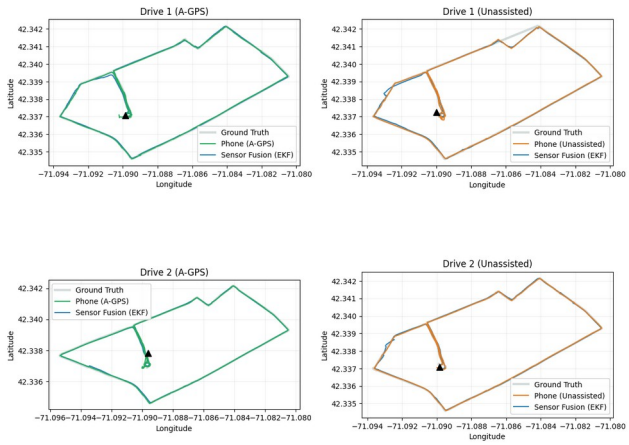
2.5 Time Synchronization and Error Metrics

While collecting driving data, it is crucial for the experimental and ground truth positions to be synchronized in order to evaluate the error. We approached this challenge in two ways. First, by spatial nearest neighbor matching, and then by UTC time synchronization. The nearest neighbors approach gives the on-road accuracy of the system, while the time-based synchronization captures lag errors in addition to the horizontal GPS error for moving data. Error is measured in meters using the haversine distance between matching experimental and ground truth coordinates. Errors less than a threshold of 5 meters are considered converged due to imperfect colocation within the vehicle and allowance for some expected GPS error due to occlusions in the city environment and the use of non-RTK GPS.

3. Driving Results

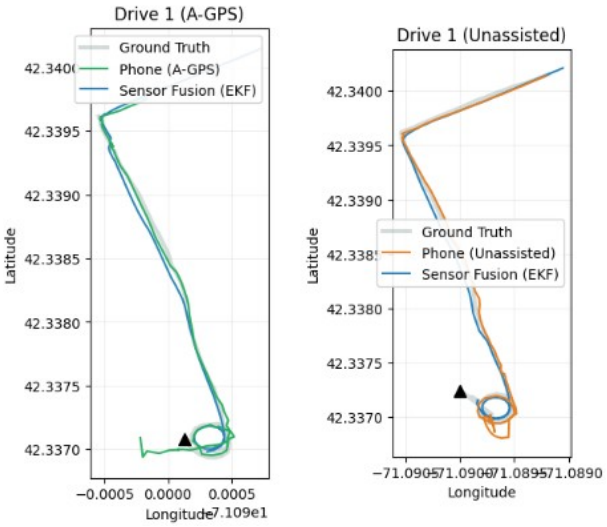
3.1 Trajectories

Figure 1 shows the full-route trajectories for all four driving trials. The results indicate that over a long collection, all methods are able to suitably estimate position.

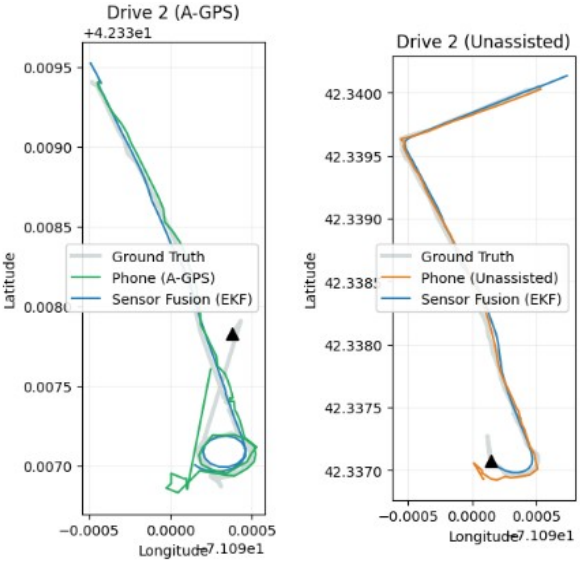


[Figure 1: Driving trajectories of Drive 1 and Drive 2 (A-GPS and Unassisted)]

Figures 2 and 3 zoom into the first 120 seconds of Drive 1. These plots reveal large first fix errors for A-GPS estimates, indicating that A-GPS does not improve first fix quality, and can actually be a detriment. The unassisted phone shows a smaller but visible first fix error, while the initial EKF estimate has minimal error. As the car continues along its path, all three measurements consistently converge to the ground truth.



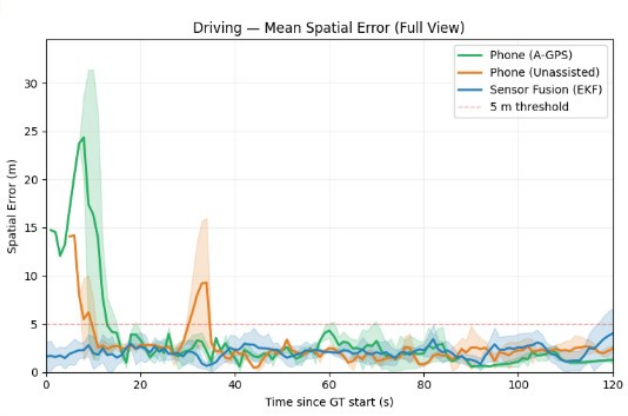
[Figure 2: Trajectories zoomed to first 120 s showing cold-start divergence (Drive 1)]



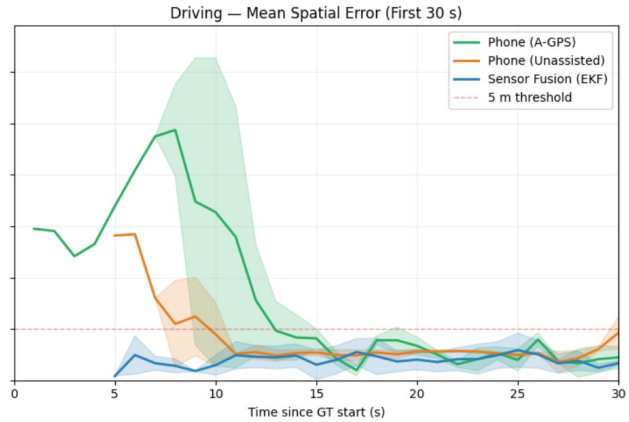
[Figure 3: Trajectories zoomed to first 120 s showing cold-start divergence (Drive 2)]

3.2 Spatial Error

Figures 4 and 5 show the mean spatial error over a 120-second and 30-second window, respectively. These errors are estimated using a nearest-neighbor approach. Assisted GPS has an early first fix but high initial error. Because of this large initial error, A-GPS estimates actually take longer on average to converge to within an acceptable error distance from the ground truth. This is consistent with the results from the trajectory plots, and indicates that the horizontal precision of A-GPS can be lower than ordinary GPS following a cold start. Because this error is calculated using spatial matching and not time-synchronization, errors caused by lag while the car is moving are not included. This error represents on-road accuracy. The EKF results are consistently within the threshold of acceptable error.



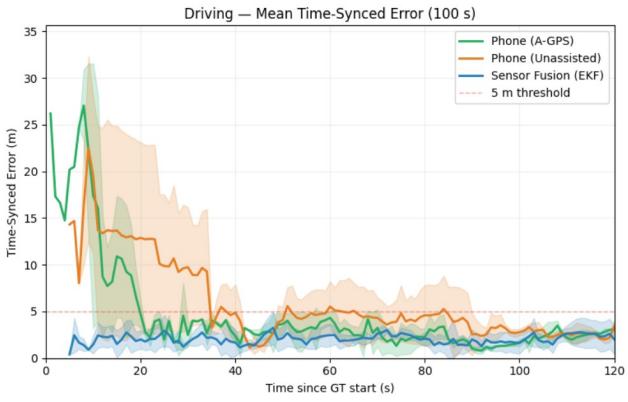
[Figure 4: Driving — Mean Spatial Error (Full View)]



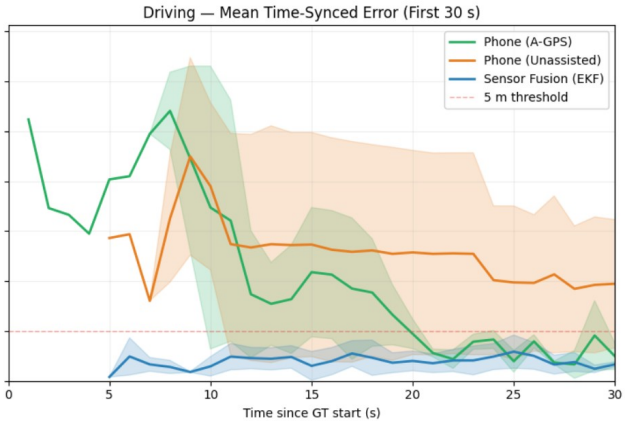
[Figure 5: Driving — Mean Spatial Error (First 30 s)]

3.3 Time-Synced Error

Figures 6 and 7 show the mean time-synced error over the full 100-second window. Unlike spatial matching, this metric matches points by absolute timestamp, so it captures whether each system is at the right place at the right time. A key difference from the spatial error plot is that the unassisted phone stays elevated longer here (from 10 to 30 seconds), suggesting it is not only spatially offset but also temporally delayed relative to ground truth. So A-GPS gets quicker results early on, but is less spatially accurate. Eventually, both varieties of phone GPS converge near the ground truth, indicating that this temporal lag for the phone's unassisted GPS is a transient effect of the cold-start. EKF has consistently low error.



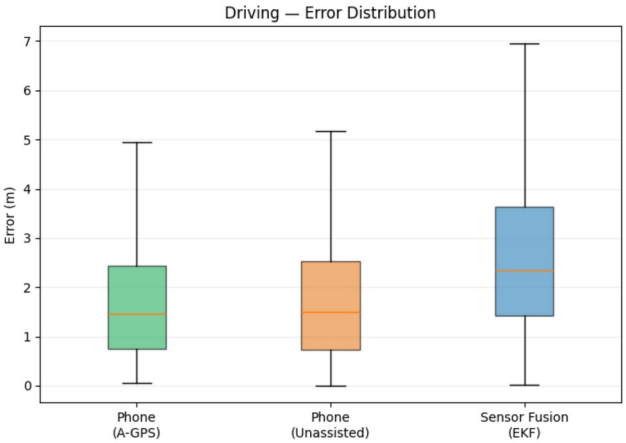
[Figure 6: Driving — Mean Time-Synced Error (100 s)]



[Figure 7: Driving — Mean Time-Synced Error (First 30 s)]

3.4 Overall Error Distribution

Figure 8 shows the overall error distribution across all driving trials. The two phone conditions have nearly identical medians and interquartile ranges, both around 1.5 m, with the upper range reaching roughly 5 m. The EKF box is wider with a higher median around 2.3 m and an upper range extending to about 7 m. This indicates that all three measurements are similarly accurate once converged, but that the custom EKF may have slightly higher variance due to less stringent tuning and calibration of the Kalman filter. All three have mean error well within the threshold of expectation.



[Figure 8: Driving — Error Distribution]

3.5 Driving Trial Results

Tables 1 and 2 summarize the mean error, standard deviation, and first-fix error for both the phone and EKF across the first 120 seconds of all four trials. The EKF maintains the lowest and most consistent mean error across all trials. Both phone conditions achieve similar mean errors, but the A-GPS phone shows a substantially higher spatial standard deviation, driven by the large first-fix spike. The first-fix column indicates inconsistent A-GPS first-fix accuracy, confirming that first-fix accuracy is highly variable and not reliably improved by A-GPS

assistance.

Trial	Mean Ph	Std Ph	Mean EKF	Std EKF	First Ph	First EKF
Drive 1 (A-GPS)	2.7m	4.9m	2.2m	1.2m	28.8m	0.9m
Drive 1 (Unassisted)	2.6m	2.5m	2.1m	1.4m	1.3m	1.0m
Drive 2 (A-GPS)	3.5m	4.2m	2.2m	1.0m	2.5m	4.2m
Drive 2 (Unassisted)	2.5m	2.2m	1.8m	1.2m	14.1m	0.4m

[Table 1: Driving Trial Results — Spatial Matching]

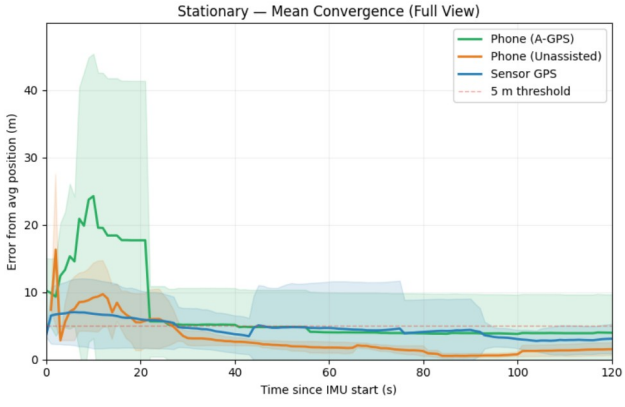
Trial	Mean Phone	Std Phone	Mean EKF	Std EKF	1st Phone	1st EKF
Drive 1 (A-GPS)	4.0m	5.8m	3.9m	2.1m	30.9m	5.0m
Drive 1 (Unassisted)	9.5m	8.0m	3.9m	2.5m	23.0m	5.8m
Drive 2 (A-GPS)	5.2m	6.9m	2.9m	1.7m	58.2m	11.3m
Drive 2 (Unassisted)	2.7m	2.4m	2.4m	1.4m	14.3m	3.0m

[Table 2: Driving Trial Results — Time-synced]

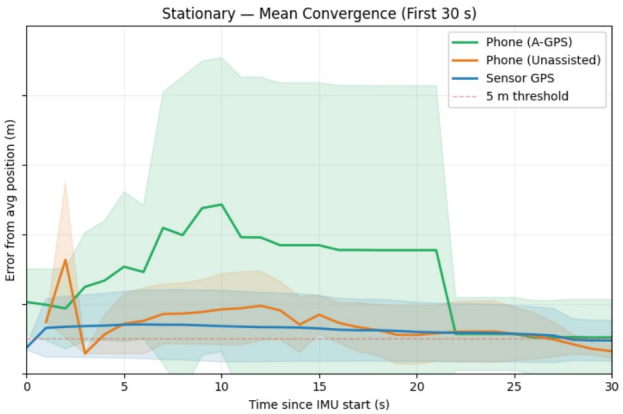
4. Stationary Results

4.1 Convergence Error

Figures 9 and 10 show the mean error for positional estimates in an unmoving location over 120-second and 30-second windows. In this case, ground truth was not collected separately. Instead, error is measured against the mean position over the full three-minute collection window for each measurement. The results hold up to the observations from our driving experiment, with high initial A-GPS error followed by eventual convergence among all systems. In this experiment, we did not use an EKF for our custom system but instead relied on raw GPS estimates due to the lack of relevant IMU data for a fixed sensor.



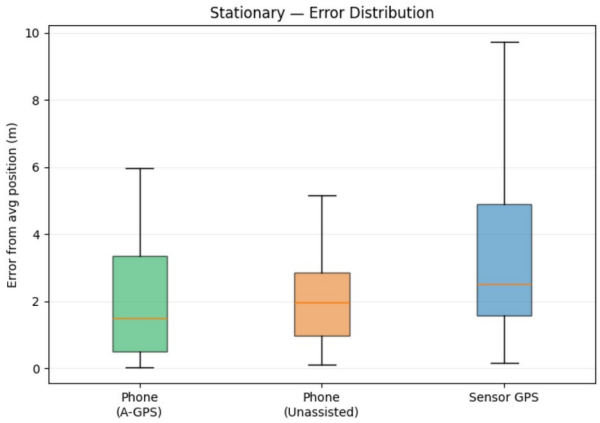
[Figure 9: Stationary — Mean Convergence (Full View)]



[Figure 10: Stationary — Mean Convergence (First 30 s)]

4.2 Overall Error Distribution

Figure 11 shows the stationary error distribution. The unassisted phone has the most compact box with a median of around 2 m and an upper range reaching roughly 5 m. The A-GPS phone has a similar median but wider spread, reaching about 6 m. The Sensor GPS has the largest spread, with an upper range reaching nearly 10 m.



[Figure 11: Stationary — Error Distribution]

4.3 Stationary Trial Results

Table 2 summarizes the mean error and standard deviation for both the phone and GPS across both conditions. The most striking finding is the A-GPS phone's standard deviation, which is nearly three times higher than that of the unassisted phone. The A-GPS phone's mean error is also almost double the unassisted phone's mean error. This confirms that large transient spikes in the A-GPS trials cause significant inaccuracies, even though steady-state accuracy across both conditions is comparable.

Condition	Phone	Standard Deviation for the Phone	GPS	Standard Deviation for the GPS
A-GPS	4.9m	10.6m	3.7m	2.6m
Unassisted	2.7m	3.4m	2.7m	1.8m

[Table 3: Stationary Trial Results]

5. Limitation and Challenges

Several limitations should be acknowledged. The ground truth iPhone carries an estimated 2–5 m of position uncertainty in open sky, which sets a floor on the resolution of all error comparisons. The iPhone's proprietary internal fusion algorithms cannot be decoupled from the A-GPS contribution, meaning we cannot fully isolate the effect of A-GPS from the phone's internal processing.

A key methodological challenge was time synchronization. The phyphox iPhone logs and ROS2 bag files use independent clocks, and on loop routes, time-based matching failed because the vehicle passes the same location twice. We resolved this by switching to spatial nearest-neighbor matching and using time-synced error as a secondary metric.

A second challenge was EKF drift. The VN-100 accelerometer output includes gravity, and without removing it, the position error accumulated to hundreds of meters within seconds. We resolved this by calibrating gravity from the first 100 IMU samples at the start of each trial and subtracting it before integration.

Finally, the small number of driving trials, two per condition, which limits the statistical power of our comparisons, and all trials were conducted in relatively open-sky suburban conditions. Results may differ in urban canyons, tunnels, or areas with heavy signal obstruction.

6. Conclusion and Future Work

We conclude that, while Assisted-GPS helps with initial first fix timing, it has mixed results for accuracy in the time period following that first. Our results indicate that the initial cellular estimate can be extremely inaccurate, causing convergence to true position estimates to be just as slow or slower than the same device achieves without GPS assistance. However, while unassisted GPS is more spatially accurate during the convergence period, it does suffer more lag. This causes elevated error in unassisted GPS when using time-synced driving measurements, but not when measuring spatial driving error or stationary convergence error.

We also found that the quality of the GPS hardware is the main contributor to first-fix quality, with the standalone BU-353S4 GPS consistently outperforming the iPhone’s small, internal GPS module (whose hardware is not publicly disclosed). However, we did find that the iPhone positional outputs had lower variance over an extended run, perhaps suggesting that Apple’s proprietary sensor fusion algorithm compensates for their hardware in the long run.

Future work in this domain might include testing how these results differ depending on environmental factors such as cellular reception and GPS availability, testing with warm starts in addition to cold starts, and comparing to positional estimates using sophisticated fusion methods such as UKF or a particle filter.

7. References

[1] D. Titterton and J. Weston, Strapdown Inertial Navigation Technology, 2nd ed. Reston, VA: AIAA, 2004.

[2] F. Caron, E. Duflos, D. Pomorski, and P. Vanheeghe, "GPS/IMU data fusion using multisensor Kalman filtering: introduction of contextual aspects," Information Fusion, vol. 7, no. 2, pp. 221–230, 2006.

[3] K. Merry and P. Bettinger, "Smartphone GPS accuracy study in an urban environment," PLOS ONE, vol. 14, no. 7, e0219890, 2019.